



DEVELOPMENT LOG

Architecture & Systems Overhaul

Branch 4.7 | 23-27 August 2026

Prepared from repository history and the architecture handoff

Scope: 0e3d6450 through 6cb4fbc6 | Unity 6000.3.22f1

Executive Summary

AUGUST 2026

Between 23 and 27 August 2026, JunkLite's branch 4.7 was substantially modernized to make combat, abilities, spawning, cameras, UI, and scene transitions more reliable and easier to extend: damage now flows through one request/result contract, player and enemy responsibilities are cleanly separated, weapons and status effects report actual applied damage, mods run through per-instance executions with deterministic cleanup and composable player locks, and the monolithic game-manager path was divided into a persistent Game Root, scene-local Level Context, PlayerLifecycle, and GameUIManager. The V2.5 scene received a repeatable migration and validation workflow, cameras now rebind across spawn and respawn, restart loading overlaps video playback with asynchronous scene loading, duplicate managers no longer destroy shared roots, and the obsolete dash VFX was removed. The refactor compiles in Unity 6000.3.22f1, the focused damage suite passes 7/7, and the redesigned scene validator passes; final confidence now depends primarily on the documented Play Mode checks for animation, physics, VFX, camera, HUD, pause, death, and respawn behavior.

DEVELOPMENT SNAPSHOT

7

COMMITTS

89

FILES CHANGED

9.2k

LINES ADDED

7/7

DAMAGE TESTS

WHAT CHANGED AT A GLANCE

- Combat foundation: unified damage requests/results, accurate applied-damage reporting, and clearer player/enemy ownership.
- Abilities and mods: per-slot runtime execution, explicit install/combat lifecycles, cancellation cleanup, and composable locks.
- Runtime architecture: persistent Game Root, scene-local Level Context, dedicated player lifecycle, and dedicated game UI lifecycle.
- Scene experience: robust camera rebinding, safer duplicate handling, faster restart transitions, and a validated V2.5 migration tool.
- Quality: focused EditMode coverage, source-level legacy cleanup, build verification, documentation, and removal of obsolete dash effects.

Note: the large deletion count is dominated by serialized V2.5 scene cleanup and rebuilding, not feature removal.

Architecture at a Glance

CLEAR OWNERSHIP, SMALL BOUNDARIES

RUNTIME BOUNDARY

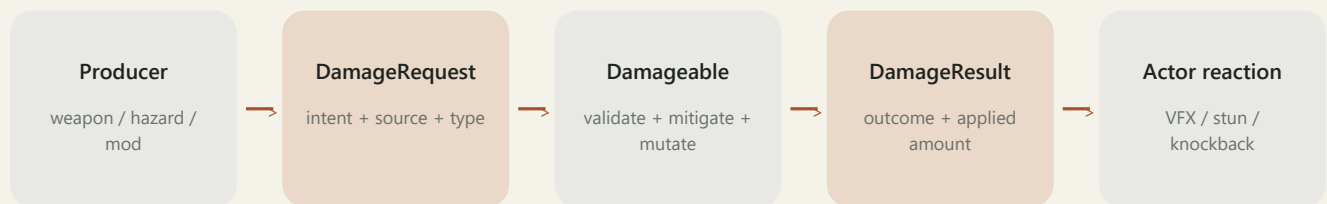
Persistent Game Root

Survives scene changes and owns GameManager, PlayerLifecycle, GameUIManager, GameInputManager, PlayerCombatTracker, the runtime gameplay canvas, and the event system.

Scene-local Level Context

Defines level identity, whether a player should spawn, and typed player spawn points. Camera rigs remain scene-local because they belong to authored level content.

DAMAGE FLOW



DESIGN PRINCIPLES NOW ENFORCED

- Composition for shared capabilities; no universal character hierarchy.
- One authoritative health-mutation path for all first-party damage.
- Actors own reactions; damage resolution owns validation and math.
- ScriptableObjects remain configuration, not mutable execution state.
- Every long-running ability has one owner and one cleanup path.
- Managers split only at concrete lifecycle or scene boundaries.

Combat & Character Systems

AUTHORITATIVE DAMAGE, CLEAR REACTIONS

1 / DAMAGE CONTRACTS

One request, one result, one health mutation path

DamageContracts.cs introduces DamageRequest, DamageOutcome, DamageResult, IDamageReceiver, and hierarchy-aware receiver resolution. Damageable now validates the request, applies armor mitigation, mutates health through AttributeManager, returns clamped applied damage, and emits resolution events only when damage is actually applied.

- Removed the legacy IDamageable, DamageInfo, boolean TakeDamage path, conversion helpers, fallback resolution, and the old damage event.
- Outcomes explicitly distinguish applied, blocked, parried, invulnerable, friendly-fire, dead, and invalid requests.
- Generic hit-stun was removed from Damageable; player and enemy code own presentation and interruption behavior.

2 / PLAYER AND ENEMY SEPARATION

Different actors keep different state machines

PlayerCharacter

Now derives directly from MonoBehaviour and owns player input orchestration, parry, immunity, shields, damage reactions, death and respawn, grabbing, and presentation. Defense order is explicit and deterministic.

EnemyBase

A deliberately small enemy-only base owns stats, damage binding, health/death, healing, activation, and forced death. Targeting, movement, AI, interruption, knockback, drops, and presentation stay enemy-specific.

3 / WEAPON AND HIT HANDLING

- WeaponManager now uses DamageRequest/DamageResult for melee, blast, hitscan, piercing, single-target, and multi-target attacks.
- OnEnemyHit publishes actual applied damage; hit VFX, hit-stop, recoil, melee durability, and hit events require a successful result.
- Piercing and area attacks deduplicate resolved receivers, preventing multi-collider enemies from receiving repeated hits.
- Ranged durability is still consumed on fire, preserving the intended weapon-cost behavior.

Mods, Abilities & Status Effects

PER-INSTANCE EXECUTION WITH DETERMINISTIC CLEANUP

EXECUTION ARCHITECTURE

Runtime state moved out of shared assets

Mod definitions remain ScriptableObject configuration, while ModInstance owns per-slot durability, charges, cooldown, and execution state. A player-owned ModExecutionRunner starts each activation through a ModExecutionContext, which owns cancellation callbacks and player-control scopes. Leaving combat mode, disabling the player, or removing a mod now cancels active execution before teardown.

LIFECYCLE AND CONTROL SAFETY

Explicit lifecycle

OnInstalled / OnRemoved now describe slot ownership. OnCombatModeEntered / OnCombatModeExited describe temporary availability. Mode changes no longer pretend to equip or unequip an item.

Composable locks

Disposable input and damage-immunity leases allow multiple abilities to overlap safely. Movement, physics overrides, and rigidbody state restore only after the final owner releases its scope.

ABILITY AND STATUS CORRECTIONS

Pulse Barrier

Shield/VFX state is runtime-owned; cleanup covers expiry, removal, disable, and cancellation.

Don't Blink

Targeting and strike execution use result-based damage; rejected hits do not produce false success feedback.

Social Distance

AOE targets are deduplicated and feedback only appears for applied results.

Energy Wave

Cooldown is no longer started twice; capture occurs only after applied damage and prior physics/NavMesh state is restored.

Phantom Strike

Charge reset follows damage-resolution events; slam AOE deduplicates targets and execution/camera cleanup is centralized.

Fire / Electric

Status effects preserve the player as source; electric chaining deduplicates multi-collider enemies.

Status ticks

Events publish actual applied damage and reuse an expiry buffer instead of allocating a new list every frame.

Game Flow & Runtime Infrastructure

SPAWN, UI, AND SCENE OWNERSHIP

PERSISTENT ROOT AND SCENE CONTEXT

A reusable foundation for every level

- Game Root is duplicate-safe and persistent; Level Context is standalone and scene-local.
- The V2.5 rebuild command removes obsolete bootstrap, input, and UI infrastructure while preserving authored geometry, cameras, enemies, pickups, audio, projectiles, drops, and feedback services.
- The migration command is manual-only and saves plus validates the rebuilt scene in one reviewed operation.

LIFECYCLE EXTRACTIONS

PlayerLifecycle

Owns the player prefab, current-player identity, typed spawn selection, death observation, lifecycle events, and delayed soft respawn. Direct consumers no longer call player APIs on GameManager. Spawn events drive camera, HUD, combat tracking, trigger state, enemies, and debug UI rebinding.

GameUIManager

Owns HUD, pause, game-over, and loading prefabs and instances. It binds to game state and player spawn, shows HUD only for a live player in an enabled level, closes inventory before pause, and restores HUD binding after soft respawn.

RELIABILITY FIXES

- Duplicate GameManager, GameManager, and PlayerCombatTracker instances now remove only the duplicate component, never their shared host object.
- This closes an Awake-order race that could destroy the complete Game Root before V2.5 player spawning.
- GameManager is smaller and now concentrates on global state, pause coordination, scene transitions, and music.
- Compatibility fields remain only as bridges for older scenes that have not yet migrated to the reusable prefab.

Camera, Loading & Project Cleanup

PLAYER EXPERIENCE AND MAINTENANCE

CAMERA OWNERSHIP

Every registered camera follows the current player

- CameraManager remains scene-local and subscribes directly to PlayerLifecycle spawn events.
- Main, spawn, death, and configured level cameras share one cached, deduplicated registry and rebind on spawn and respawn.
- CameraSwitchTrigger registers and binds newly selected cameras on demand; follow-freeze state survives switching.
- Respawn prioritizes the configured spawn camera, then main, then the first registered fallback.
- The V2.5 validator requires one scene-local CameraManager, one CinemachineBrain, and a main camera from the scene rig.

RESTART AND LOADING TRANSITION

Loading now happens in parallel with the video

The previous restart path waited for the full loading video before beginning LoadSceneAsync and then held activation behind a serialized six-second debug delay. The new path starts both operations together, waits only until the scene is ready and the video is complete, removes the obsolete delay from manager prefabs, and keeps input disabled until scene references, player, camera, and UI are initialized.

EDITOR, ASSETS, AND CLEANUP

- Project metadata was updated for Unity 6000.3.22f1, including generated Input System actions, package manifests/locks, and solution metadata.
- The obsolete dash afterimage scripts and their metadata were removed, completing the old-dash optimization cleanup.
- Manager prefabs, selected UI scripts/materials, and V2.5 serialized data were updated to match the new lifecycle ownership.
- ARCHITECTURE_HANDOFF.md now records implemented boundaries, validation evidence, known gaps, and the next safe work order.

Verification & Current Status

EVIDENCE RECORDED IN UNITY 6000.3.22F1

RECORDED VERIFICATION

- Runtime and editor assemblies compiled with 0 errors after the mod, spawn, camera, lifecycle, UI, and migration-tool changes.
- DamagePipelineTests passed 7/7 in EditMode, covering applied versus requested damage, rejection outcomes, immunity, death/revive, initialization, and lock composition.
- PlayerLifecycleConfigurationTests passed 3/3 before UI extraction; one added reusable-root UI configuration test still awaits an in-Editor rerun.
- Tools > JunkLite > V2.5 > Validate Redesigned Setup passed with one configured PlayerLifecycle hosted by Game Root.
- A source search found no remaining first-party references to the removed legacy damage contracts or events.

SYSTEM STATUS

AREA	STATUS	READINESS NOTE
Damage pipeline	IMPLEMENTED	Legacy path removed; focused tests pass.
Player / enemy split	IMPLEMENTED	Representative gameplay verification pending.
Weapons and statuses	IMPLEMENTED	Verify hit feedback and timing in Play Mode.
Mods and ability locks	IMPLEMENTED	Cancellation, physics, and VFX checks pending.
Game Root / V2.5	VALIDATED	Scene validator passes; visual approval pending.
Camera and respawn	IMPLEMENTED	Repeated cross-scene Play Mode run pending.
UI lifecycle	IMPLEMENTED	One added configuration test and Play Mode flow pending.
Manager cleanup	IN PROGRESS	Legacy fallbacks remain for unmigrated scenes.

Build note: the wider project still reports pre-existing analyzer, obsolete-API, and unused-field warnings. These were not introduced as blocking compile errors by this refactor.

Playtest Gate & Next Steps

WHAT REMAINS BEFORE GAMEPLAY-CLOSED

PRIORITY PLAY MODE CHECKS

- Combat: representative player, enemy, melee, ranged, piercing, status, lifesteal, and pogo interactions.
- Mods: enter/exit combat mode, unequip during execution, last-durability activation, player disable/death, and full cleanup for Pulse Barrier, Don't Blink, Social Distance, Energy Wave, and Phantom Strike.
- Camera: initial follow, death switch, respawn snap, trigger switching, freeze/unfreeze, zoom, and repeated V2.5 loading from another scene.
- UI: exactly one HUD/pause/game-over/loading instance, inventory-before-pause, death presentation, restart, HUD rebinding, and no player HUD when SpawnPlayer is disabled.
- Loading: confirm immediate async start, no former six-second freeze, clean activation, and restored input/camera/UI.
- Inspector: verify representative player, enemy, weapon, status, and mod prefabs; then perform one final serialized-reference search before deleting CharacterBase.cs.

RECOMMENDED SEQUENCE

01

Play-test the completed combat/mod slice

Fix regressions inside the implemented boundaries before adding new abstractions.

02

Harden the Game Root migration

Rebuild only when needed, validate V2.5, then migrate one gameplay scene and one menu scene with review.

03

Verify player and UI lifecycles

Confirm spawn, death, soft respawn, pause, HUD, camera, combat tracking, and scene transitions as one flow.

04

Review WeaponManager responsibilities

Extract only a proven boundary; equipment/loadout/pickup ownership is the first likely candidate.

DEFINITION OF READY

The architecture slice is structurally complete. It should be called gameplay-closed only after the Play Mode checklist passes and representative prefab references are verified in the Inspector.

Commit-by-Commit Development Log

SOURCE WINDOW: 0E3D6450..6CB4FBC6

23 AUG		0e3d6450	Rashid Hafez
		Optimization: removed the obsolete dash implementation and afterimage scripts.	
25 AUG		bd53e3ae	Ajask1122
		Updated Unity/editor-generated project files, Input System actions, packages, and solution metadata.	
25 AUG		6248f976	Ajask1122
		Added the architecture handoff and project constraints as the refactor source of truth.	
25 AUG		46f3c34e	Jazz
		Introduced damage contracts, player/enemy separation, Game Root, Level Context, V2.5 migration tooling, and focused tests.	
26 AUG		fceecea0	Ajask1122
		Completed spawn/camera work, migrated weapons and status effects, and implemented per-instance mod execution, cleanup, and player locks.	
26 AUG		f035e7f3	Ajask1122
		Extracted PlayerLifecycle, rebuilt V2.5 scene infrastructure, and reduced GameManager responsibilities.	
27 AUG		6cb4fbc6	Jazz
		Extracted GameUIManager, completed UI/player lifecycle integrations, strengthened migration validation, and added configuration tests.	

SOURCE & SCOPE

This devlog summarizes the clean working tree on branch 4.7 using Git history and diff statistics from bd35410e to 6cb4fbc6, plus ARCHITECTURE_HANDOFF.md for implementation and verification detail. The window contains 7 commits, 89 changed files, 9,183 additions, and 42,876 deletions. Large scene-file churn is described as infrastructure cleanup rather than gameplay removal.